

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 05

String Concatenation

1) Concatenate First and Last Name

Description: Write a Java program that prompts the user to input their first and last name separately and then concatenates them into a full name.

Hint:

- Use the + operator or concat() method to join the first and last names with a space in between.

2) Create a Sentence Using Multiple Strings

Description: Write a Java program that takes three different strings as input and concatenates them to form a complete sentence.

Hint:

- Use the + operator to concatenate the strings with appropriate spaces and punctuation.

Adding Numbers and Strings

3) Sum of Two Numbers as Strings

Description: Write a Java program that prompts the user to input two numbers as strings and then adds them together as integers.

Hint:

- Convert the string inputs to integers using Integer.parseInt() and then perform the addition.

4) Concatenate Numbers as Strings

Description: Write a Java program that takes two numbers as strings and concatenates them instead of adding them.

Hint:

- Directly use the + operator to concatenate the string representations of the numbers.

Strings - Special Characters

5) Print a Sentence with Special Characters

Description: Write a Java program that prints a sentence containing special characters like double quotes, new lines, and tabs.

Hint:

- Use escape sequences like `\`, `\n`, and `\t` to include special characters in the string.

6) Format a Multi-line Address

Description: Write a Java program that takes user input for street, city, and zip code, and formats it as a multi-line address using special characters.

Hint:

- Use `\n` for new lines and `\t` for indentation if needed.

Java Math

7) Calculate the Square Root of a Number

Description: Write a Java program that prompts the user to input a number and then calculates and prints the square root of that number.

Hint:

- Use the `Math.sqrt()` method to calculate the square root.

8) Generate a Random Number Between 1 and 100

Description: Write a Java program that generates and prints a random number between 1 and 100.

Hint:

- Use `Math.random()` and manipulate the result to fall within the desired range.

Java Booleans

9) Check if a Number is Greater than 100

Description: Write a Java program that prompts the user to input a number and checks if the number is greater than 100.

Hint:

- Use a boolean variable to store the result of the comparison and print the result.

10) Check if Two Strings are Equal

Description: Write a Java program that prompts the user to input two strings and checks if they are equal.

Hint:

- Use the equals() method to compare the strings and store the result in a boolean variable.

Java Type Casting

11) Convert a Double to an Integer

Description: Write a Java program that prompts the user to input a decimal number (double) and then converts it to an integer.

Hint:

- Use type casting (int) to convert the double to an integer.

12) Cast a Larger Data Type to a Smaller Data Type

Description: Write a Java program that prompts the user to input a large integer value (e.g., long) and then converts it to a smaller data type (e.g., byte).

Hint:

- Use type casting (byte) to convert the long to a byte and observe the result.